

# Mathematical Models at the Population Level to Support Immunotherapy Design

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June 5, 2026

## 1 Introduction

The immune system plays a crucial role in defending the body against diseases, including cancer, by identifying and eliminating abnormal cells. Among its most effective components are T cells, which specifically recognize and attack malignant targets. However, under conditions of chronic stimulation like persistent exposure to tumor associated antigens, these T cells can enter a dysfunctional state known as immune exhaustion.

Exhausted T cells are immune cells that have become too weak to fully eliminate the threat. This may result in a localized stalemate, where the tumor is contained but not completely erased. Exhaustion is not a uniform binary state: T cells exist along a continuous spectrum, ranging from fully functional to deeply exhausted. Understanding this heterogeneity is really important because even highly exhausted cells maintain some functional baseline, which means that there are possibilities for immunotherapeutic interventions.

## 2 Immune system functioning

In a common scenario, the immune system follows a precise scheme in the fight against antigens:

Naïve T cells that find the antigen begin their first stage of reproduction and interpret the incoming signals from the environment in order to differentiate into effector cells. We can distinguish two types of effector cells, i.e. CD4+ Helper T cells and CD8+ Cytotoxic T cell (CTLs): the first ones have a coordinating role by sending chemical messages in the form of cytokines, while CTLs are the one who actually kill the infected cells. Once the threat has been neutralized, most of the effector cells start the apoptosis process through Regulatory T cells. The remaining ones will differentiate into Memory T cells that can reactivate in case of a second infection of the same pathogen.

During chronic infections, such as cancer, this scheme may fail: in this case, there is an overactivation of T cells and the healthy effector cells start to become exhausted: their efficiency against the tumor starts to drastically reduce and their death rate increases, leading to a localized stalemate against the infection. In this scenario, the tumor stays at a consistent size until, because of his own nature, it develops and mutates in order to avoid the immunosuppression and starts growing again.

### 3 Formulation of the Model

The foundational model made by Simmons and Levy [1] describes the development of cellular exhaustion and the resulting tumor-immune stalemate. The system of ordinary differential equations (ODEs) is formulated as follows:

$$\begin{cases} \dot{N}(t) &= \alpha N \left(1 - \frac{N}{T_c}\right) - h \left(\frac{\lambda_K NK + \lambda_E NE}{N + K + E + 1}\right) \\ \dot{K}(t) &= hPK - \delta_K K \\ \dot{B}(t) &= 2^m s_B N + hPB - r_E B - \delta_B B \\ \dot{E}(t) &= r_E B - \delta_E E \\ \dot{P}(t) &= \rho_K K + \rho_B B - hP(K + B) - \delta_P P \end{cases} \quad (1)$$

Here, the probability of contact between different populations is modeled as a product term and the state variables represent four cellular population and one cytokine.

More specifically, the individual terms in the equations represent the following biological mechanisms:

- **Tumor dynamics ( $\dot{N}$ ):** The term  $\alpha N \left(1 - \frac{N}{T_c}\right)$  models the logistic growth of the tumor, where  $\alpha$  is the baseline growth rate and  $T_c$  is the maximum carrying capacity of the tissue. The negative term represents the tumor killing. Here,  $\lambda_K$  and  $\lambda_E$  define the specific killing abilities of healthy and exhausted cells, respectively. The denominator  $(N + K + E + 1)$  acts as a saturation function to model the physical competition and interactions in the tumor environment.
- **Healthy CTLs ( $\dot{K}$ ):** The term  $hPK$  describes the proliferation of healthy cells, which is stimulated by the IL-2 cytokine. The term  $\delta_K K$  represents their natural death rate.
- **Progenitor Exhausted T cells ( $\dot{B}$ ):** The term  $2^m s_B N$  represents the supply of new progenitor cells, which is directly dependent on the tumor size. Similar to healthy cells,  $hPB$  models their proliferation driven by IL-2. The term  $r_E B$  represents the differentiation rate at which these cells turn into terminally exhausted cells, while  $\delta_B B$  is their death rate.
- **Terminally Exhausted T cells ( $\dot{E}$ ):** The term  $r_E B$  is the positive influx of cells differentiating from the progenitor stage. Since they are fully exhausted, they miss a proliferation term and simply die at a rate of  $\delta_E E$ .
- **IL-2 Cytokine ( $\dot{P}$ ):** The terms  $\rho_K K$  and  $\rho_B B$  describe the production of IL-2 by healthy and progenitor cells, respectively. The negative term  $hP(K + B)$  accounts for the consumption of the cytokine by these cells during their proliferation process, and  $\delta_P P$  represents the natural decay rate of the cytokine over time.

## 4 Numerical simulations

### 4.1 Data reproduction

Using the parameters from the original study [1], the model was simulated numerically. As shown in the figures below, there is an initial phase of tumor growth, which drives the expansion of both Progenitor ( $B$ ) and Terminally Exhausted ( $E$ ) cells. Meanwhile, the healthy CTLs ( $K$ ) decline over time due to low IL-2 levels. Furthermore, a localized tumor-immune stalemate is reached: the tumor burden stabilizes at a non-zero value, indicating a chronic failure of the immune system.

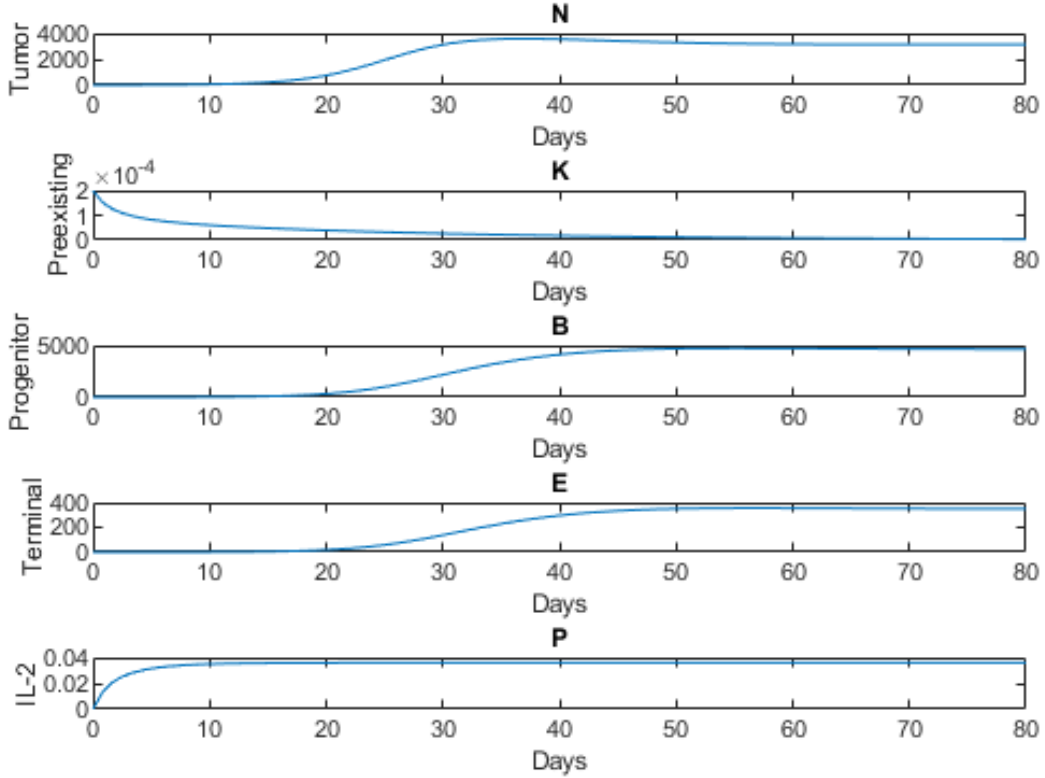


Figure 1: Reproduction of the homogeneous model dynamics showing the evolution of  $N$ ,  $K$ ,  $B$ ,  $E$ , and  $P$  over time.

### 4.2 Extending the model: multiple exhaustion states

To capture the heterogeneity of T cell exhaustion, the  $E$  group of cells has been divided following different levels of exhaustion:

- $E_0$ : Initially Exhausted T cells (functional, capable of tumor killing).
- $E_{1/2}$ : Middle-exhausted T cells (partially dysfunctional).
- $E_1$ : Deeply-exhausted T cells (fully exhausted, negligible killing capabilities).

The new model is the following:

$$\begin{cases} \dot{N}(t) = \alpha N \left(1 - \frac{N}{T_C}\right) - h \left( \frac{\lambda_k NK + \lambda_0 NE_0 + \lambda_{1/2} NE_{1/2} + \lambda_1 NE_1}{N + K + E_0 + E_{1/2} + E_1} \right), \\ \dot{K}(t) = hPK - \delta_k K, \\ \dot{B}(t) = 2^m S_B N + hPB - r_E B - \delta_B B, \\ \dot{E}_0(t) = r_E B - r_t E_0 - \delta_0 E_0, \\ \dot{E}_{1/2}(t) = r_t E_0 - r_t E_{1/2} - \delta_{1/2} E_{1/2}, \\ \dot{E}_1(t) = r_t E_{1/2} - \delta_1 E_1 \\ \dot{P}(t) = \rho_K K + \rho_B B - hP(K + B) - \delta_P P, \end{cases} \quad (2)$$

where:

- $\delta_0$ ,  $\delta_{1/2}$ , and  $\delta_1$  represent the death rates for the initially, middle and deeply exhausted T cells, respectively. Deeper exhaustion is linked with a shorter lifespan, therefore, these rates follow the inequality  $\delta_0 < \delta_{1/2} < \delta_1$ .
- $\lambda_0$ ,  $\lambda_{1/2}$ , and  $\lambda_1$  represent the tumor killing rates for each corresponding subpopulation. Because progressive exhaustion severely influences effector function, the killing capacity decreases as cells transition keep going, leading to  $\lambda_0 > \lambda_{1/2} > \lambda_1$ .
- $r_t$  is the phenotypic transition rate, in other words, the speed at which T cells progressively pass from one state to the other in the following order:  $E_0 \rightarrow E_{1/2} \rightarrow E_1$ .

#### 4.2.1 Equivalence under Homogeneous Assumptions

First, we demonstrate that if we assume uniform killing and death rates ( $\lambda_0 = \lambda_{1/2} = \lambda_1 = \lambda_E$  and  $\delta_0 = \delta_{1/2} = \delta_1 = \delta_E$ ), the extended system turns back to the original model. In fact, if we define the total exhausted population as  $E = E_0 + E_{1/2} + E_1$  and substitute the respective differential equations, we can see that:

$$\begin{aligned} \dot{E} &= \dot{E}_0 + \dot{E}_{1/2} + \dot{E}_1 \\ \dot{E} &= (r_E B - r_t E_0 - \delta_E E_0) + (r_t E_0 - r_t E_{1/2} - \delta_E E_{1/2}) + (r_t E_{1/2} - \delta_E E_1) \\ \dot{E} &= r_E B - \delta_E (E_0 + E_{1/2} + E_1) \\ \dot{E} &= r_E B - \delta_E E. \end{aligned} \quad (3)$$

By canceling the intermediate transition terms ( $r_t E_0$  and  $r_t E_{1/2}$ ), the final result matches exactly the equation for  $\dot{E}(t)$  in the original system. Furthermore, if we consider also the

equation for  $\dot{N}$ , we got:

$$\begin{aligned}
\dot{N} &= \alpha N \left( 1 - \frac{N}{T_C} \right) - h \left( \frac{\lambda_k N K + \lambda_0 N E_0 + \lambda_{1/2} N E_{1/2} + \lambda_1 N E_1}{N + K + E_0 + E_{1/2} + E_1 + 1} \right) \\
&= \alpha N \left( 1 - \frac{N}{T_C} \right) - h \left( \frac{\lambda_k N K + \lambda_E N E_0 + \lambda_E N E_{1/2} + \lambda_E N E_1}{N + K + E_0 + E_{1/2} + E_1 + 1} \right) \\
&= \alpha N \left( 1 - \frac{N}{T_C} \right) - h \left( \frac{\lambda_k N K + \lambda_E N (E_0 + E_{1/2} + E_1)}{N + K + (E_0 + E_{1/2} + E_1) + 1} \right) \\
&= \alpha N \left( 1 - \frac{N}{T_C} \right) - h \left( \frac{\lambda_k N K + \lambda_E N E}{N + K + E + 1} \right)
\end{aligned} \tag{4}$$

confirming that, with homogeneous assumptions, the model obtained is again the original one. This equivalence is validated numerically below.

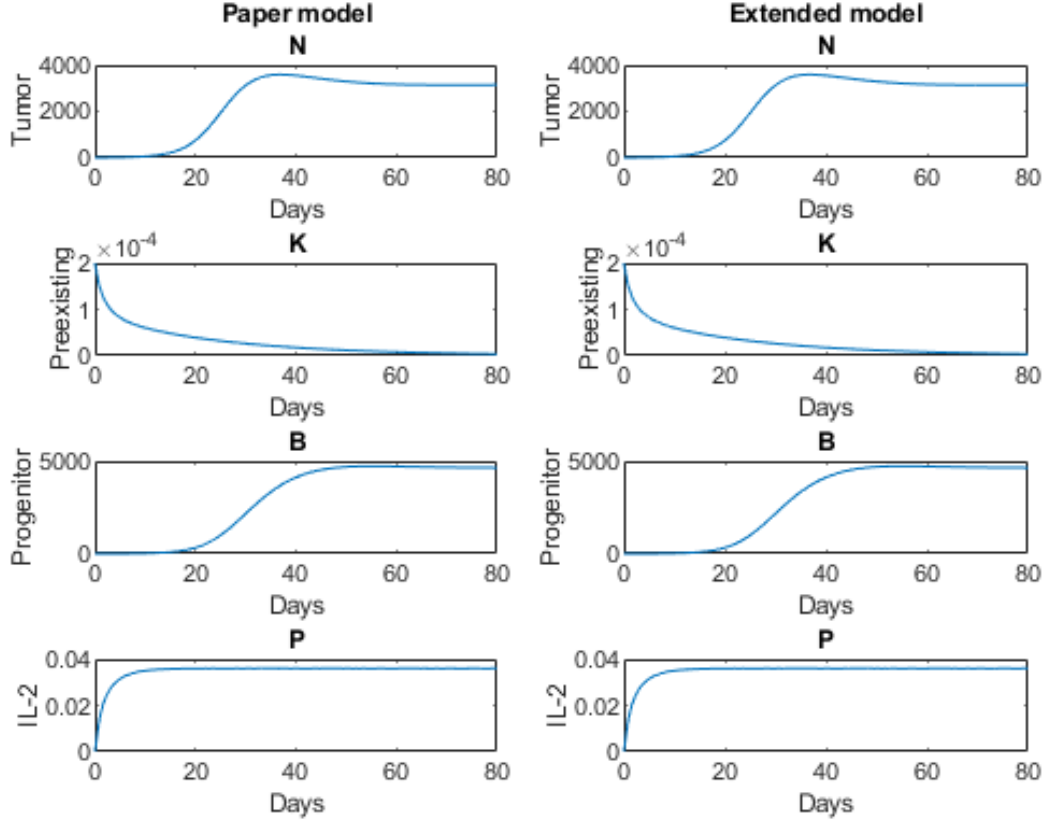


Figure 2: Comparison between the original system and the new one with  $r_t = 5r_E$

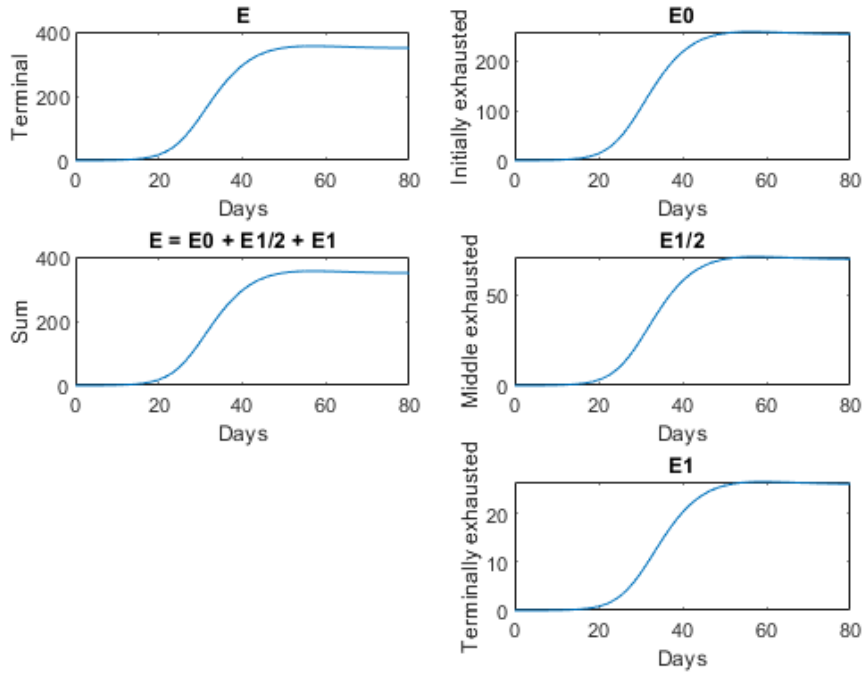


Figure 3: Comparison with  $r_t = 5r_E$  showing that the sum of the extended compartments ( $E_0 + E_{1/2} + E_1$ ) perfectly overlaps with the  $E$  compartment from the homogeneous paper model.

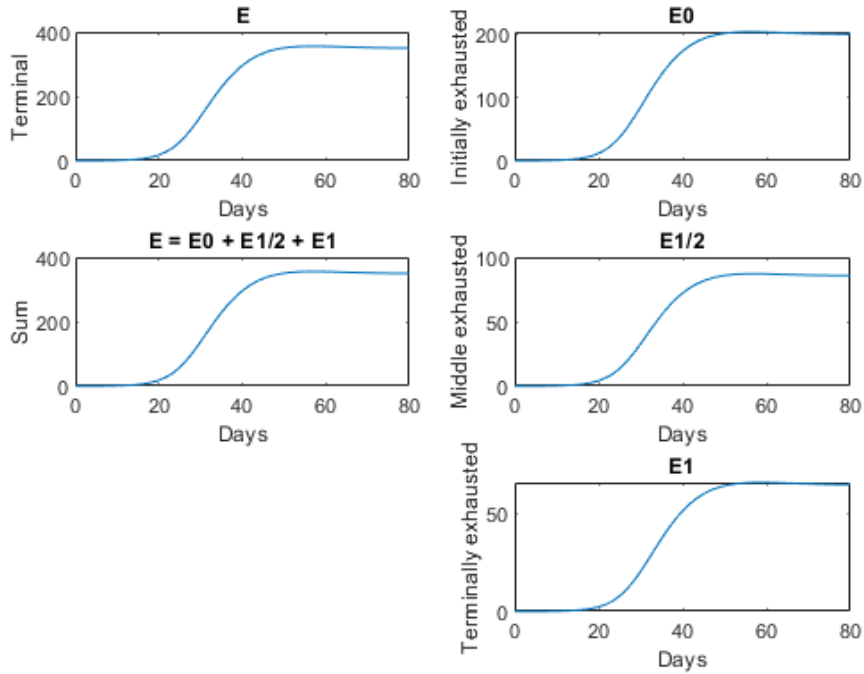


Figure 4: Comparison with  $r_t = 10r_E$  showing that while the sum of the extended compartments ( $E_0 + E_{1/2} + E_1$ ) perfectly overlaps with the  $E$  compartment, its composition is different.

It is good to notice that while the sum  $E_0 + E_{1/2} + E_1$  remains the same with both values of  $r_t$ , its composition is different:

- When  $r_t$  is lower ( $r_t = 5r_E$ , Figure 3), the transition is slower, causing most cells to remain in an initial exhaustion state ( $E_0$ ).
- When  $r_t$  is doubled ( $r_t = 10r_E$ , Figure 4), cells become exhausted much faster. As a result, the peak  $E_0$  population drops, while the deeply exhausted groups ( $E_{1/2}$  and  $E_1$ ) grow more.

#### 4.2.2 Introduction of heterogeneity in killing and death rates

Deeply exhausted T cells have shorter lifespans and lower tumor-killing capacities. To simulate that, we define a continuous linear function for the killing rate  $\lambda(y)$  and death rate  $\delta(y)$  as functions of  $y \in [0, 1]$ :

$$\lambda(y) = \lambda_K + y(\lambda_E - \lambda_K)$$

$$\delta(y) = \delta_K + y(\delta_E - \delta_K)$$

Using these linear functions, we evaluate the intermediate state ( $y = 1/2$ ) to derive  $\lambda_{1/2}$  and  $\delta_{1/2}$ :

$$\lambda_{1/2} = \lambda_K + \frac{1}{2}(\lambda_E - \lambda_K)$$

$$\delta_{1/2} = \delta_K + \frac{1}{2}(\delta_E - \delta_K)$$

With this choice, we assumed that Initially Exhausted T cells ( $y = 0$ ) have the same killing/death rate as Healthy T cells and that Deeply Exhausted T cells ( $y = 1$ ) have the same killing/death rate as the Terminally T cells.

### 4.3 Simulation and Analysis of the Extended Model

The implementation of these heterogeneous rates modify the dynamics of the interaction, as shown below.

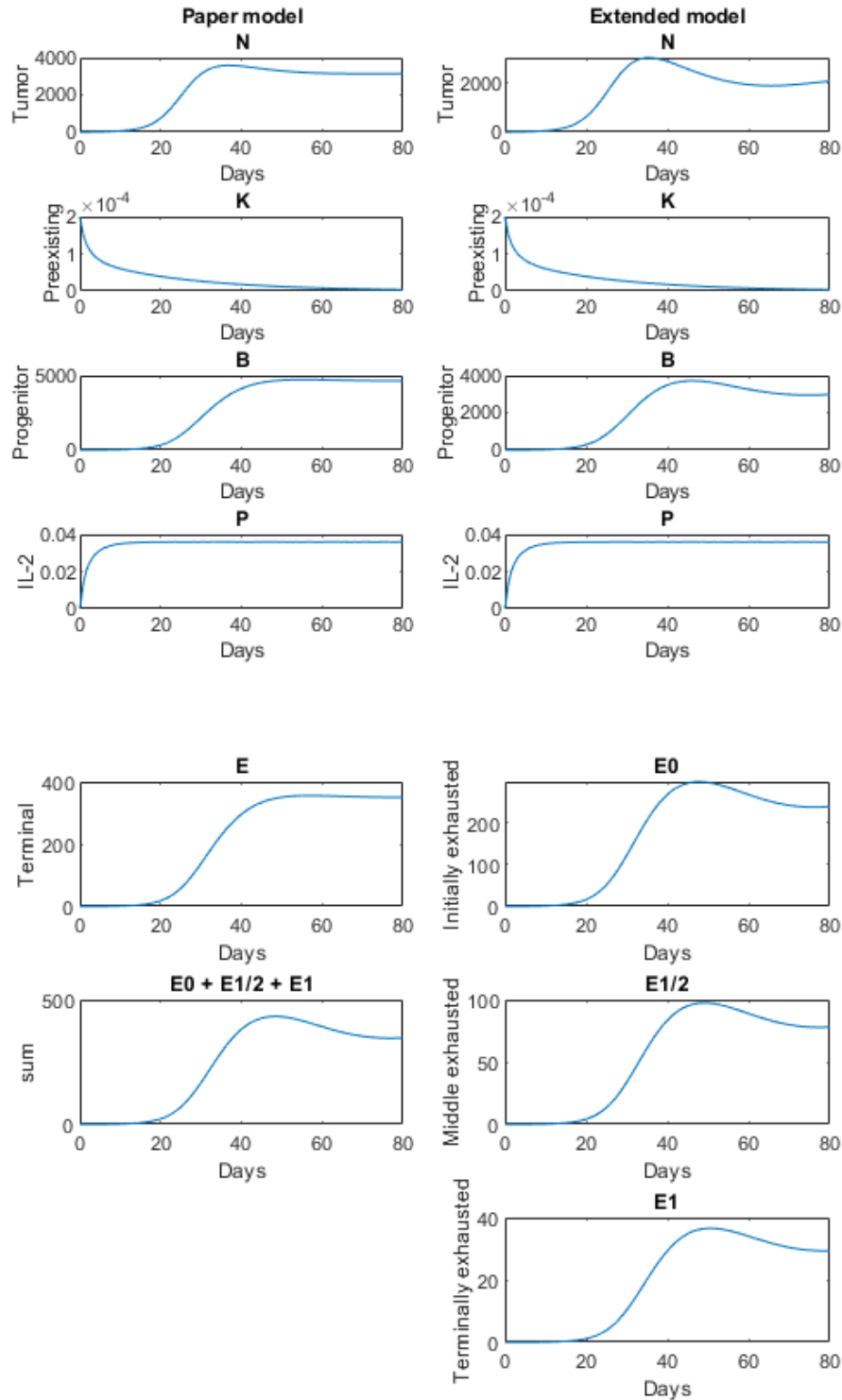


Figure 5: Evolution of variables under heterogeneous assumptions compared to the original system with  $r_t = 5 \cdot r_E$ .



By adding specific death and killing rates based on exhaustion level, the early exhausted cells ( $E_0$ ) provide a strong and immediate killing power that is similar to fully functional CTLs. However, because they transform into deeply exhausted cells ( $E_1$ ) and have higher death rates, the size and stability of the tumor-immune balance changes.

Specifically, the early attack from  $E_0$  causes a sharp drop in tumor size. This creates a new balance at a much lower level than what the homogeneous model predicts. This drop causes a major "domino" effect on the other cell types. Because the presence of Progenitor cells ( $B$ ) depends directly on the tumor size through the developmental program term  $2^m s_B N$ , a smaller tumor means less antigen stimulation.

Consequently, the Progenitor population ( $B$ ) reaches a peak and then drops to a lower steady state than in the homogeneous model. This smaller supply limits the growth of  $E_0$ . As a result, fewer cells move on to the later  $E_{1/2}$  and  $E_1$  stages. This explains why the total number of exhausted cells ( $E_0 + E_{1/2} + E_1$ ) drops to a lower balance after its first higher spike.

In conclusion, having more functional cells ( $E_0$ ) clears the tumor faster. However, by removing the constant antigen stimulation, the immune system cuts off its own supply of T cells, resulting in another permanent stalemate.

## 5 Continuous Population Model

Biological phenomena involving phenotypic variation can often be more accurately modeled as continuous spectra rather than discrete compartments. In this section we will formulate an extension of the model introducing a continuous phenotype  $y \in [0, 1]$ . We will define  $E(t, y)$  as a density function representing the total number of exhausted cells at time  $t$  characterized by phenotype  $y$ , governed by a Partial Differential Equation.

### 5.1 Derivation of the model

We want to distinguish exhausted cells based on their exhaustion level, represented by a continuous phenotype variable  $y \in [0, 1]$ . To model how the characteristics of these cells change as they become more exhausted, we define the tumor-killing parameter  $\lambda_t(y)$  and the death rate  $\delta_t(y)$  as functions of the phenotype  $y$ :

$$\begin{aligned}\lambda_t(y) &= \lambda_K + y(\lambda_E - \lambda_K) \\ \delta_t(y) &= \delta_K + y(\delta_E - \delta_K).\end{aligned}\tag{5}$$

Starting from the original model equation:

$$\frac{dE_{tot}(t)}{dt} = r_E B(t) - \delta_E E_{tot}(t)\tag{6}$$

we define a density function  $E(t, y)$  such that the total number of cells is expressed by:

$$E_{tot}(t) = \int_0^1 E(t, y) dy.\tag{7}$$

If we consider a small piece of the space  $\Delta y$  at time  $t$ , we have that the number of cells in this piece is defined by:

- number of still functional cells,
- number of cells that mutate towards exhaustion,
- number of dying cells, regulated by  $\delta_t(y)$ .

Introducing  $v(t, y)$  as the cells aging speed, the equation becomes:

$$\frac{\partial}{\partial t}[E(t, y)\Delta y] = v(t, y)E(t, y) - v(t, y + \Delta y)E(t, y + \Delta y) - \delta_t(y)E(t, y)\Delta y\tag{8}$$

and dividing by  $\Delta y$ , we obtain the incremental ratio:

$$\frac{\partial}{\partial t}E(t, y) = \frac{v(t, y)E(t, y) - v(t, y + \Delta y)E(t, y + \Delta y)}{\Delta y} - \delta_t(y)E(t, y).\tag{9}$$

Taking the limit  $\Delta y \rightarrow 0$ , we obtain:

$$\frac{\partial}{\partial t}E(t, y) = -\frac{\partial}{\partial y}(v(t, y)E(t, y)) - \delta_t(y)E(t, y)\tag{10}$$

rearranging the terms, we recognize an advection-reaction PDE:

$$\frac{\partial}{\partial t}E(t, y) + \frac{\partial}{\partial y}(v(t, y)E(t, y)) = -\delta_t(y)E(t, y). \quad (11)$$

To have a unique solution we have to put some boundary conditions, and we choose them in order to respect the initial equation: at the beginning of exhaustion, we know that the exhausted cells come from the term  $r_E B(t)$ , that we can consider as the incoming flux on the left side, i.e.

$$v(t, 0)E(t, 0) = r_E B(t). \quad (12)$$

The final system of ODEs and PDEs is the following:

$$\begin{cases} \dot{N}(t) = \alpha N \left(1 - \frac{N}{T_c}\right) - h \left( \frac{\lambda_K N K + N \int_0^1 \lambda_t(y) E(t, y) dy}{N + K + E_{tot} + 1} \right) \\ \dot{K}(t) = h P K - \delta_K K \\ \dot{B}(t) = 2^m s_B N + h P B - r_E B - \delta_B B \\ \dot{P}(t) = \rho_K K + \rho_B B - h P (K + B) - \delta_P P \\ \frac{\partial E(t, y)}{\partial t} = -\frac{\partial}{\partial y}(v(t, y)E(t, y)) - \delta_t(y)E(t, y) \\ v(t, 0)E(t, 0) = r_E B(t) \end{cases} \quad (13)$$

## 5.2 Numerical approach

To numerically solve the coupled system of ordinary and partial differential equations (ODEs and PDEs), we used the Method of Lines (MOL). This technique allows to decouple the spatial and temporal discretizations: the continuous phenotype domain,  $y \in [0, 1]$ , is divided into an evenly spaced grid of  $r = 1001$  nodes with a constant spatial step size  $h$ . Consequently, the PDE governing the density of exhausted cells,  $E(t, y)$ , is transformed into a system of  $r$  interconnected ODEs.

To discretize the space, we use a first-order backward finite difference approximation (upwind scheme) for the advection term:

$$\frac{d}{dt}E_i = \frac{v_{i-1}E_{i-1} - v_i E_i}{h} - \delta_i E_i \quad \text{for } i = 1, \dots, r \quad (14)$$

This is because cells travel in only one direction in the phenotype space, meaning that they can only go towards exhaustion and cannot "heal" and reverse the aging process. The reaction term, which represents the phenotype dependent death rate  $\delta_t(y)E(t, y)$ , is calculated directly at each point on the grid. This turns the PDE into a sparse matrix,  $A_{PDE}$ , where the advection and reaction values are on the main diagonal and the subdiagonal below.

Furthermore, the boundary condition  $v(t, 0)E(t, 0) = r_E B(t)$  representing progenitor cells turning into initially exhausted cells is added by linking the  $B(t)$  variable to the first point of the grid:

$$\frac{d}{dt}E_1 = \frac{[r_E B(t)] - v_1 E_1}{h} - \delta_1 E_1 \quad (15)$$

In our numerical model, the state vector is organized as  $X = [N, K, B, P, E_1, \dots, E_r]^T$ . To correctly align the  $A_{PDE}$  matrix with this full state vector, we add  $r \times 4$  columns of zeros to

its left side. This makes sure that the discrete PDE only interacts with the correct variables. Specifically, the boundary input term is placed in the first row and third column of  $A_{PDE}$ . This matches the exact position of the progenitor cells variable  $B(t)$ :

$$A_{PDE}(1, 3) = \frac{r_E}{h}. \quad (16)$$

Because the system has many equations ( $4 + r$ ) and involves different time scales, the resulting ODE system becomes very stiff. To solve this, we calculate the time integration using MATLAB's ode15s solver. This tool is optimized for stiff problems and is based on Numerical Differentiation Formulas (NDF). To make the simulation faster, we provided the solver with the sparse Jacobian matrix  $J$ :

$$J = \begin{bmatrix} A_{ODE} \\ A_{PDE} \end{bmatrix} \quad (17)$$

In this matrix, the top block  $A_{ODE} \in \mathbb{R}^{4 \times (4+r)}$  describes the nonlinear interactions between the main variables ( $N$ ,  $K$ ,  $B$ ,  $P$ ) and the total exhausted population. The bottom block  $A_{PDE} \in \mathbb{R}^{r \times (4+r)}$  contains the dynamics of the exhaustion process, including the boundary condition. The overall structure of the Jacobian matrix can be seen below:

$$J = \left[ \begin{array}{cccc|cccc} \frac{\partial \dot{N}}{\partial N} & \frac{\partial \dot{N}}{\partial K} & 0 & 0 & \frac{\partial \dot{N}}{\partial E_1} & \dots & \frac{\partial \dot{N}}{\partial E_j} & \dots & \frac{\partial \dot{N}}{\partial E_r} \\ 0 & \frac{\partial \dot{K}}{\partial K} & 0 & \frac{\partial \dot{K}}{\partial P} & 0 & \dots & 0 & \dots & 0 \\ \frac{\partial \dot{B}}{\partial N} & 0 & \frac{\partial \dot{B}}{\partial B} & \frac{\partial \dot{B}}{\partial P} & 0 & \dots & 0 & \dots & 0 \\ 0 & \frac{\partial \dot{P}}{\partial K} & \frac{\partial \dot{P}}{\partial B} & \frac{\partial \dot{P}}{\partial P} & 0 & \dots & 0 & \dots & 0 \\ \hline 0 & 0 & \frac{r_E}{h} & 0 & -\left(\frac{v_1}{h} + \delta_1\right) & 0 & \dots & \dots & 0 \\ 0 & 0 & 0 & 0 & \frac{v_1}{h} & -\left(\frac{v_2}{h} + \delta_2\right) & \ddots & & \vdots \\ \vdots & \vdots & \vdots & \vdots & 0 & \ddots & \ddots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \frac{v_{r-2}}{h} & -\left(\frac{v_{r-1}}{h} + \delta_{r-1}\right) & 0 \\ 0 & 0 & 0 & 0 & 0 & \dots & 0 & \frac{v_{r-1}}{h} & -\left(\frac{v_r}{h} + \delta_r\right) \end{array} \right]$$

### 5.3 Results and discussion

Here below we can see the results for three different aging speeds:

**Interpretation for  $v(t, y) = 5r_E$ :**

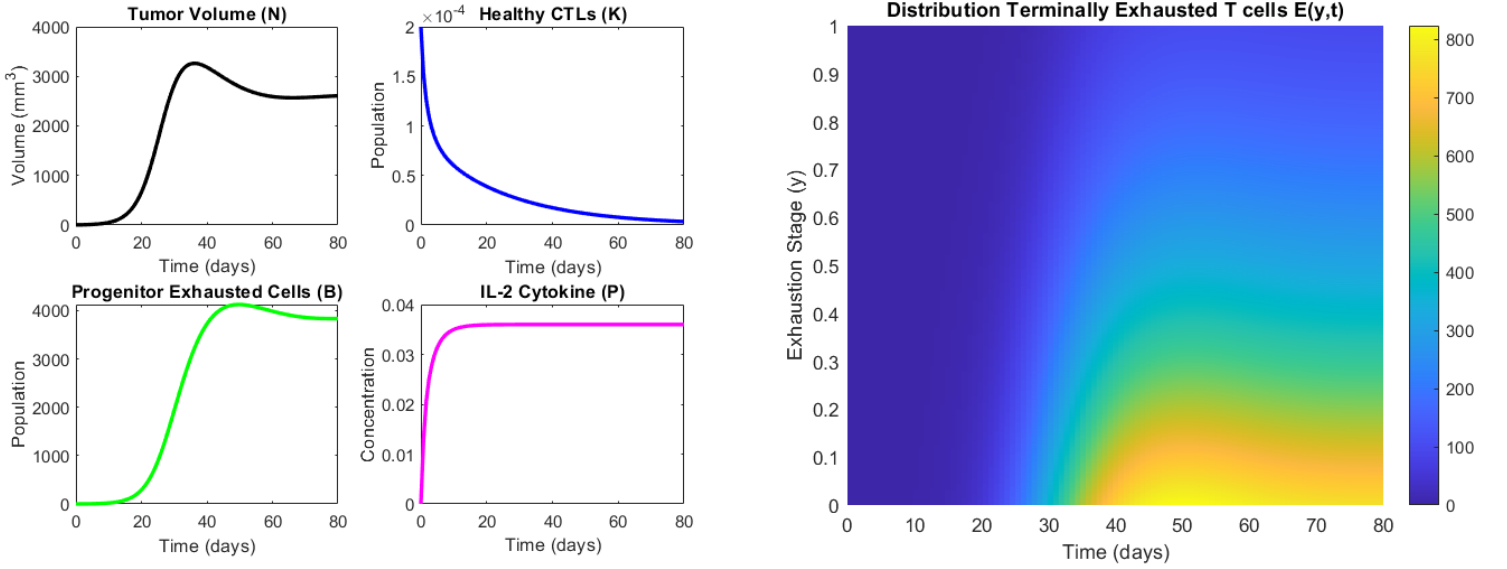


Figure 6: Results for  $v(t, y) = 5r_E$ : the left panel shows the dynamics of the main variables, while the right panel is a heatmap of the exhausted population  $E(t, y)$  over time.

Using the same speed as before, the heatmap of Terminally Exhausted T cells  $E(y, t)$  shows the distribution of the population over time. Cells start entering the continuous spectrum around day 20. This happens exactly when the progenitor cells ( $B$ ) begin to multiply much faster and, as a consequence, a greater flow of cells enters the exhaustion process.

The highest cell density, shown in yellow, is heavily concentrated in the early exhaustion stages ( $y < 0.2$ ). Looking forward, because these cells have a higher death rate based on their specific stage,  $\delta_t(y)$ , most of them die before they can age and move further along the spectrum. As a result, the deeply exhausted states ( $y > 0.5$ ) have a much lower density. This shows that, with these parameters, the exhausted population is mainly made of newly changed cells.

In this setup, the tumor stops growing at a volume of about 2700 mm<sup>3</sup> after hitting a peak near day 35 at 3400 mm<sup>3</sup>. This represents a classic stalemate between the tumor and the immune system: the system controls the tumor's fast growth, but it cannot destroy it completely.

Looking at the other main populations, this constant stimulation causes a huge increase in Progenitor Exhausted T cells ( $B$ ), following the tumor's growth and stabilizing just below 4000. On the other hand, the Healthy CTLs ( $K$ ) keep dropping toward zero. This proves that the long-term immune response depends entirely on the exhausted cells, while the fully working cells disappear.

Interpretation for  $v(t, y) = 15r_E$ :

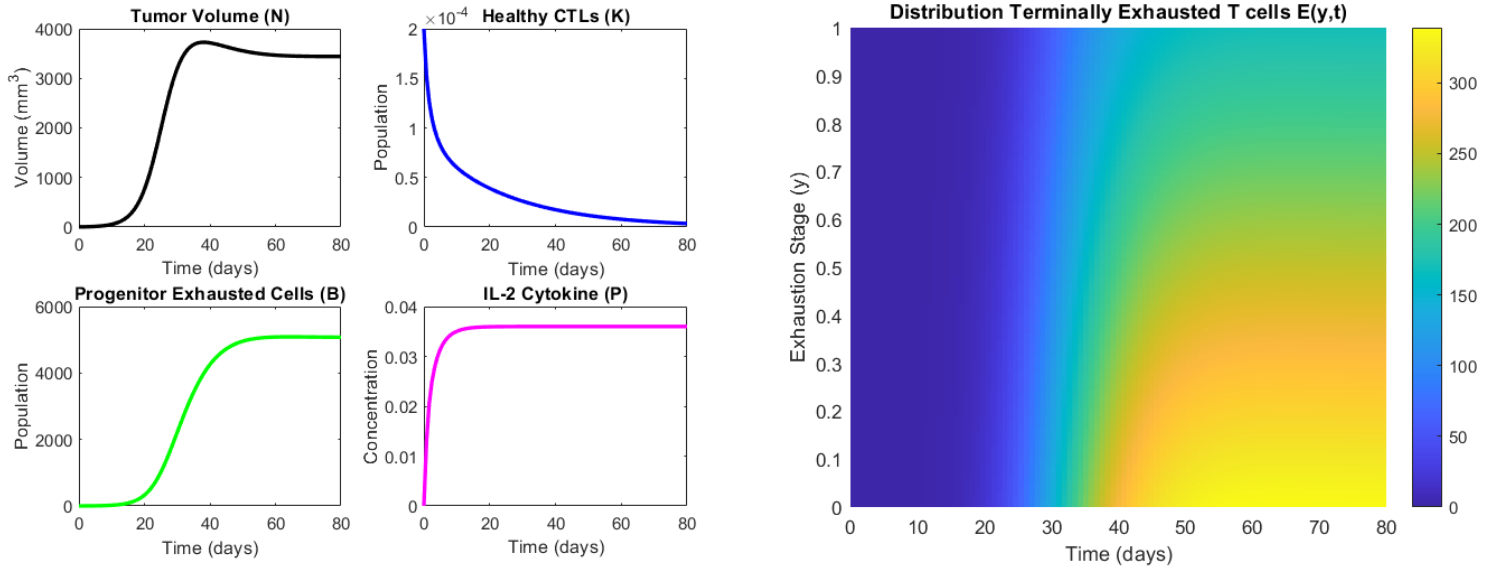


Figure 7: Results for  $v(t, y) = 15r_E$ : the left panel shows the dynamics of the main variables, while the right panel is a heatmap of the exhausted population  $E(t, y)$  over time.

In this case, we set a higher aging speed three times faster than the default. The heatmap shows that the population of exhausted cells is a lot more spread out across the spectrum and the peak density is much lower. Furthermore, since the population of cells in an advanced exhaustion state is greater and the death rate  $\delta_t(y)$  is higher for deeper exhausted cells, the total population is smaller than before. Nevertheless, a greater amount of cells reach the deeper exhaustion stages, showing that the immune system is more dysfunctional.

As a consequence, the tumor reaches his peak at 3900 mm<sup>3</sup> and stabilizes at an higher volume than before at 3500 mm<sup>3</sup>, showing that the immune system is less effective at eliminating the tumor. This much more consistent stimulation causes Progenitor Exhausted T cells (B) population to increase until it reaches around 5000 units.

Interpretation for  $v(t, y) = r_E$ :

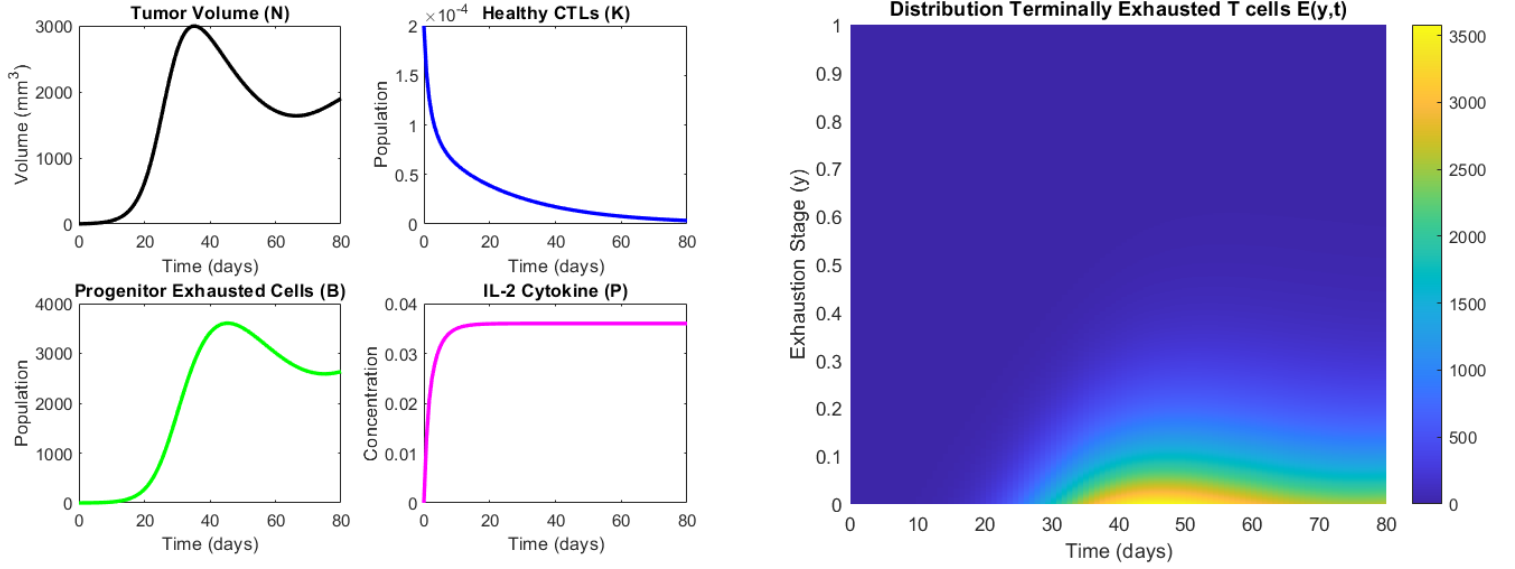


Figure 8: Results for  $v(t, y) = r_E$ : the left panel shows the dynamics of the main variables, while the right panel is a heatmap of the exhausted population  $E(t, y)$  over time.

In this case, we notice the opposite scenario: as the aging speed has been imposed to be five times slower than the default, the population of exhausted cells is mostly concentrated in a small area below  $y = 0.1$ . Here, the initial phenomenon where most cells are newly exhausted and die before they can age is even more pronounced. Another behavior that before was quite light but that now we can notice more is that going forward in time, also the density of cells in the early exhausted cells decreases. This is because the tumor reaches a much lower peak at  $3000 \text{ mm}^3$ , because of the greater killing efficiency, and then sharply decreases, causing a lower stimulation of the immune system and, consequently, a lower supply of exhausted cells. This is also reflected in the Progenitor Exhausted T cells (B) population, which reaches a much lower peak around 3600 units and then stabilizes around 2900.

Particularly interesting in this scenario is the fact that, differently from the previous cases, the tumor seems to come back increasing after day 65, instead of stabilizing at a constant level. The same behaviour seems to be presented also by the Progenitor Exhausted T cells (B) population. To better understand this phenomenon, we extended the simulation for a longer time:

### Time extended trend:

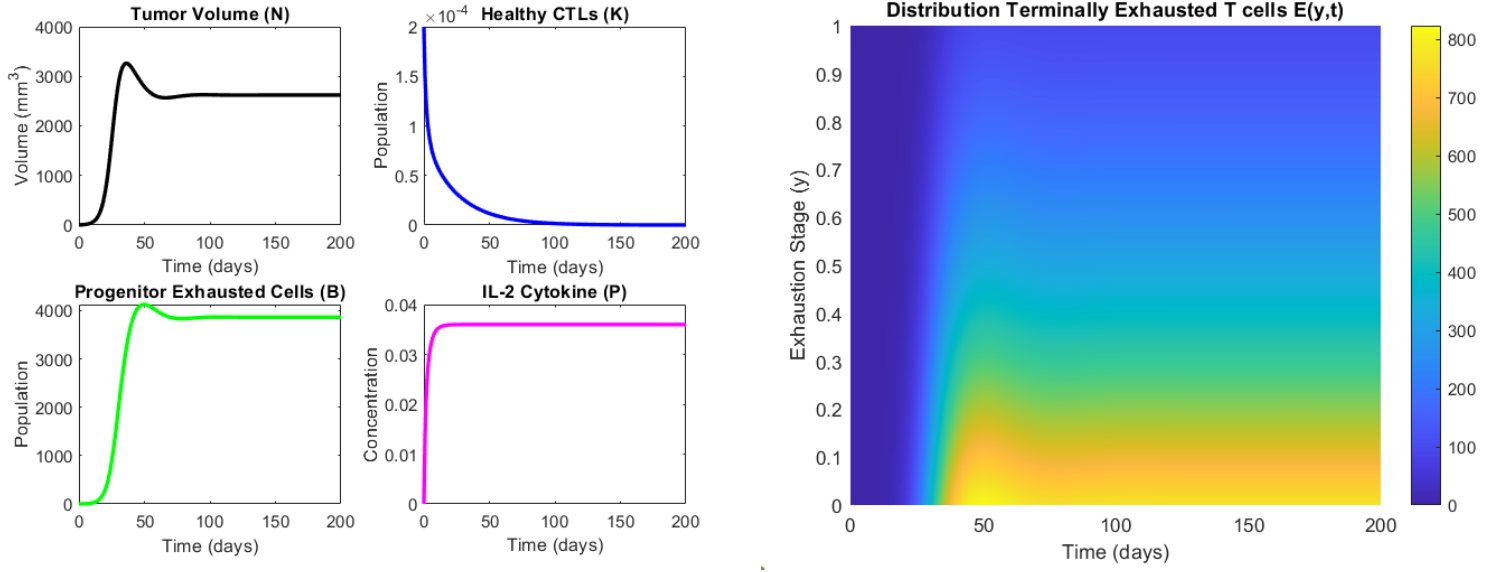


Figure 9: Results for  $v(t, y) = 5r_E$  and 200 days: the left panel shows the dynamics of the main variables, while the right panel is a heatmap of the exhausted population  $E(t, y)$  over time.

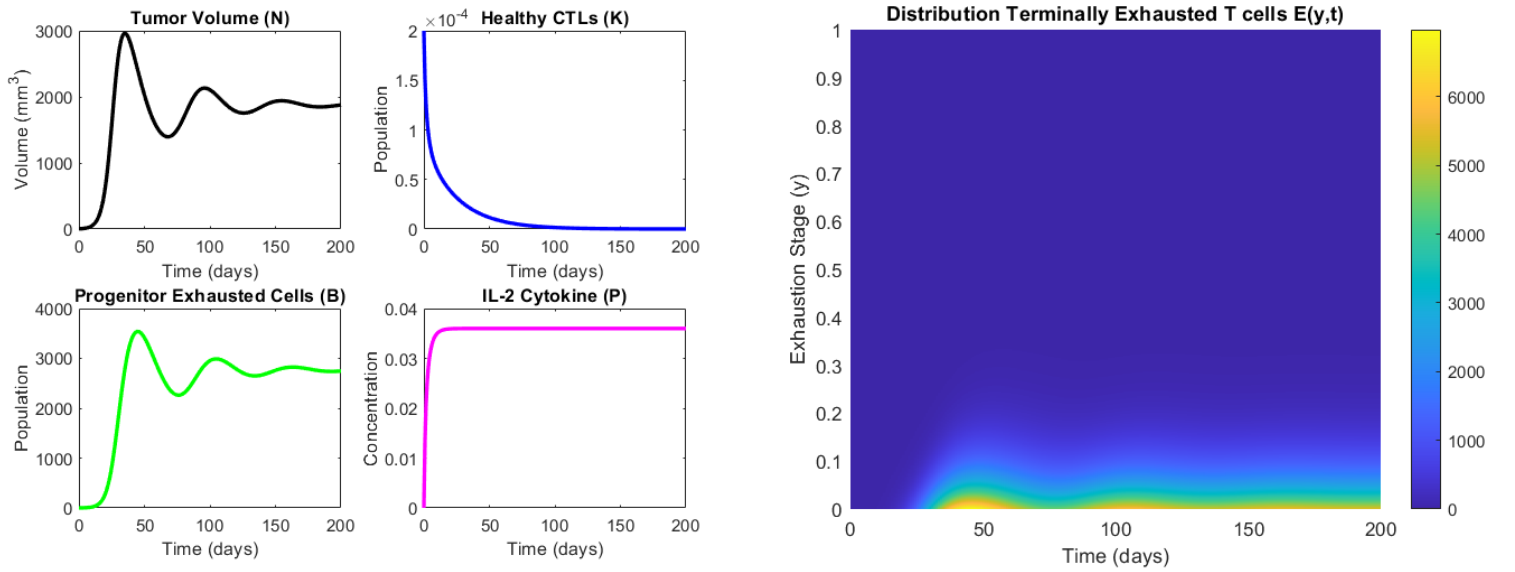


Figure 10: Results for  $v(t, y) = 0.5r_E$  and 200 days: the left panel shows the dynamics of the main variables, while the right panel is a heatmap of the exhausted population  $E(t, y)$  over time.

Exploring the dynamics for a longer time period, we can see that while with  $v(t, y) = 5r_E$  the trend follows as expected and the tumor is stable at a constant level, with  $v(t, y) = 0.5r_E$  we can clearly see the tumor and the Progenitor Exhausted T cells (B) population oscillating before converging slowly to a steady state. This perfectly represents the continuous battle between the tumor and the immune system. Initially, the tumor grows, providing a strong



antigen signal that triggers a massive production of progenitor cells ( $B$ ). These progenitors then change into early-stage terminally exhausted cells, which still have a strong ability to kill the tumor and, since the aging speed is slow, they remain functional for a long time. When this wave of functional exhausted cells attacks, the tumor shrinks. However, a smaller tumor means a weaker antigen signal. As a result, the supply of new progenitor cells drops, causing a delayed decrease in the number of functional exhausted cells. With the immune concentration temporarily reduced, the tumor starts to grow again. This cycle of growing and shrinking continues, gradually losing strength, until both populations finally settle into a chronic tumor-immune stalemate. As shown below, extending simulation time to 600 days, the system converges to the same solution as before: a perfect stalemate where neither the tumor nor the immune system completely prevails, remaining both at high levels over time.

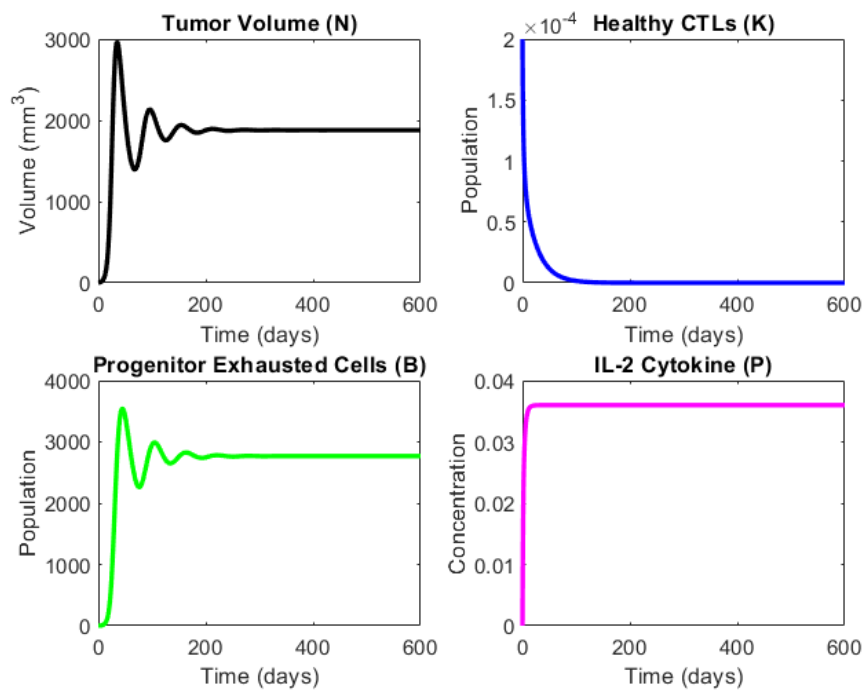


Figure 11: Results for  $v(t, y) = 0.5r_E$  and 600 days: the system slowly reaches the equilibrium.

## 5.4 Conclusions

In this study, we explored the complex dynamics between tumor growth and the immune system, focusing on the phenomenon of T cell exhaustion. Starting from a basic mathematical model, we expanded it to include multiple exhaustion stages. Finally, we developed a continuous population model using a Partial Differential Equation (PDE) to better represent the gradual aging process of exhausted T cells along a continuous spectrum.

Numerical simulations show that the speed at which T cells become exhausted ( $v(t, y)$ ) significantly affects the tumor's survival. When cells age fast, they quickly lose their ability to fight and die sooner, leading to a larger and more stable tumor volume. On the other hand, when the aging speed is slow, early stage exhausted cells remain functional for a longer time. This creates a delayed loop, showing oscillations between the tumor and the immune cells before reaching a final stalemate.

Overall, T cell exhaustion is not just a simple switch, but a dynamic and continuous process. The immune system's failure to completely destroy the tumor is partially caused by its own mechanism: as the tumor shrinks, the antigen stimulation drops, cutting off the supply of new active T cells. From a medical point of view, this mathematical model may suggest that immunotherapies could be highly effective if they target also the exhaustion speed. While alone it would not be so efficient, slowing down the transition of T cells into deeply exhausted states could help the immune system maintain a stronger and longer attack against cancer. This, coupled with increasing the killing capabilities or maintain IL-2 levels higher in time, could be a promising method to efficiently defeat chronic diseases.

## References

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- [2] The MathWorks Inc. Ordinary differential equations - matlab & simulink, 2026.